

Probabilistic Models

Probabilistic model: mathematical description of an uncertain situation

Random experiment: experiment in which the outcome varies in an unpredictable fashion, when experiment repeated under the same condition.

→ specify random experiment by stating experimental procedure and ..., what is measured

→ examples:

experiment E_1 : select a ball numbered from an urn containing balls from 1 to 50. Note the number of the ball.

experiment E_2 : toss a coin 3 times and note the # of heads.

experiment E_3 : toss a coin 3 times and note the sequence of heads and tails

Sample space: Set of all possible outcomes of an experiment denote by Ω

$$S_1 = \{1, 2, \dots, 50\}$$

$$S_2 = \{0, 1, 2, 3\}$$

$$S_3 = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

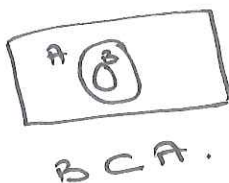
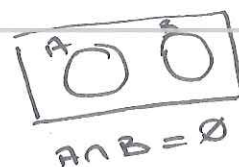
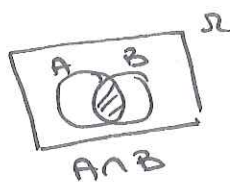
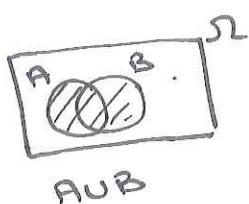
Event: collection of possible outcomes.

event E_{11} : "an even-numbered ball is selected"
 $E_{11} = \{2, 4, 6, \dots, 48, 50\}$

event E_{21} : "# of heads = # of tails"
 $E_{21} = \emptyset$

event E_{31} : "three tosses give the same outcome"
 $E_{31} = \{HHH, TTT\}$

Set operations: combine events using set operations to obtain other events. (2)



1) Commutative properties:

$$A \cup B = B \cup A, \quad A \cap B = B \cap A$$

2) Associative properties:

$$A \cup (B \cap C) = (A \cup B) \cap C, \quad A \cap (B \cup C) = (A \cap B) \cup C$$

3) Distributive properties:

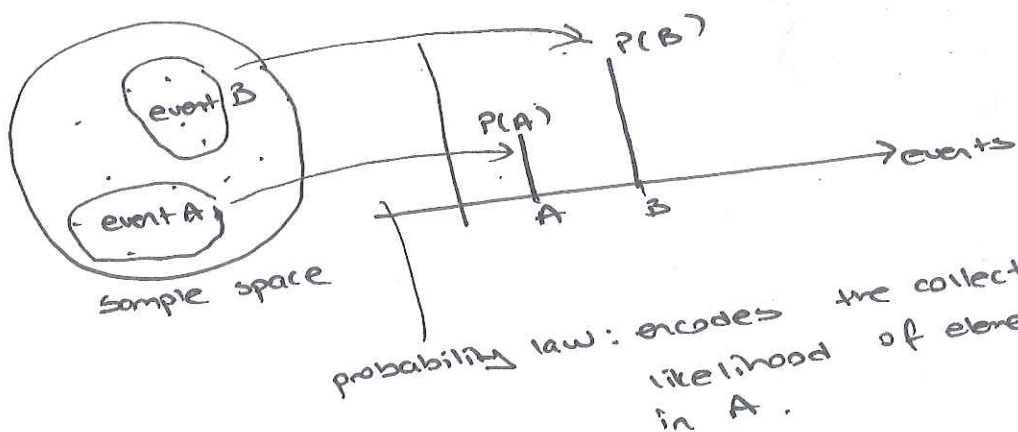
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

4) De Morgan's rules:

$$(A \cap B)^c = A^c \cup B^c, \quad (A \cup B)^c = A^c \cap B^c$$

Probability Law:

random experiment



Probability Axioms:

1) $P(A) \geq 0$ for any event A.

2) A and B disjoint events ($A \cap B = \emptyset$) $\Rightarrow P(A \cup B) = P(A) + P(B)$
more generally: $A_1, A_2, \dots$ disjoint events $\Rightarrow$
 $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

3) $P(\Omega) = 1$

Properties of Probability laws:

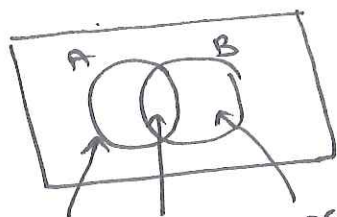
1) $P(A^c) = 1 - P(A)$.

Proof: $A \cap A^c = \emptyset$.

axiom 2: $P(A \cup A^c) = P(A) + P(A^c)$.

$\Omega = A \cup A^c \Rightarrow P(A \cup A^c) = P(\Omega) = 1 = P(A) + P(A^c)$.

2) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

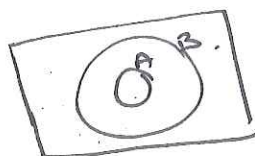


$A \cap B^c$ $A \cap B$ $A^c \cap B$ ← disjoint events

$P(A \cup B) = P(A \cap B^c) + P(A \cap B) + P(A^c \cap B)$

$\begin{cases} P(A) = P(A \cap B^c) + P(A \cap B) \\ P(B) = P(A \cap B) + P(A^c \cap B) \end{cases} \Rightarrow P(A) + P(B) = P(A \cup B) + P(A \cap B)$

3) $A \subset B \Rightarrow P(A) \leq P(B)$



$B = A \cup (A^c \cap B)$

$P(B) = P(A) + P(A^c \cap B) \geq P(A)$
 ≥ 0 .

Discrete sample spaces

Suppose sample space consists of a finite # of possible outcomes
 $\Omega = \{a_1, a_2, \dots, a_n\}$ ← all elements are mutually exclusive
 Probability of any event $B = \{a_1', a_2', \dots, a_m'\}$

$P(B) = P(\{a_1', a_2', \dots, a_m'\})$
 $= P(\{a_1'\}) + P(\{a_2'\}) + \dots + P(\{a_m'\})$

If $P(a_1) = P(a_2) = \dots = P(a_n) = \frac{1}{n}$.

$P(B) = \frac{m}{n}$ ← $\frac{\text{\# of outcomes in the event}}{\text{total \# of outcomes in sample space}}$

ex: Consider the experiment of rolling a pair of 4-sided ⁴ dice.
 $P(\{\text{sum of the rolls is even}\}) = ?$

Sample space: $\{(i,j) \mid \begin{array}{l} i: \text{result of first roll,} \\ j: \text{result of second roll} \end{array}\}$.

$$= \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4)\}$$

event: $\{(i,j) \mid \begin{array}{l} i: \text{result of first roll,} \\ j: \text{result of second roll,} \\ \text{sum of the roll is even} \end{array}\}$

$$= \{(1,1), (1,3), (2,2), (2,4), (3,1), (3,3), (4,2), (4,4)\}$$

$$P(\{\text{sum of rolls even}\}) = \frac{8}{16} = \frac{1}{2}$$

Continuous Sample Spaces:

events of interests: intervals of real line or rectangular region

ex: pick a number x at random between zero and one

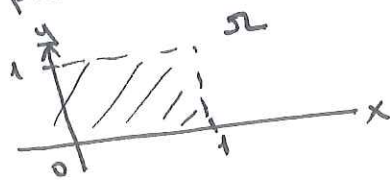
$$S = [0, 1]$$

If all outcomes s are equally likely to be selected,

$$P([0, \frac{1}{2}]) = \frac{1}{2}$$

$$P([a, b]) = b - a, \quad 0 \leq a \leq b \leq 1$$

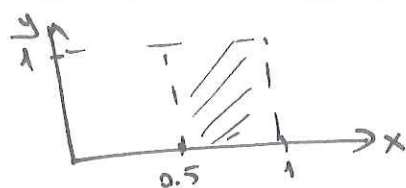
ex: pick two numbers x and y at random between 0 and 1



Assume all pairs of numbers in unit square are equally likely.

$$A = \{x > 0.5\}$$

$$P(A) = ?$$



$$P(A) = \frac{1}{2}$$

$$B = \{x > \frac{1}{2}\}$$

$$P(B) = ?$$



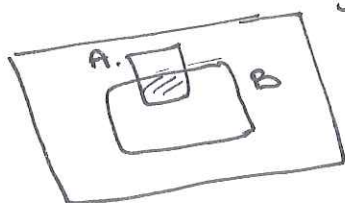
$$P(B) = \frac{1}{2}$$

Conditional Probability:

probability of an outcome in a random experiment changes if you have partial information.

(a) Probability of obtaining "2" in a roll of a die?
 $\frac{1}{6} \leftarrow S = \{1, 2, 3, 4, 5, 6\}$

(b) You know that the result is an even number; probability of obtaining "2" in a roll of a die?
 $\frac{1}{3} \leftarrow S = \{2, 4, 6\}$



Ω .

You know outcome is within B. the likelihood that outcome belongs to A?

$$\text{conditional probability of A given B?}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

↑

probability law on a new universe B.

If all possible outcomes are finitely many and equally likely:

$$P(A|B) = \frac{\# \text{ of elements in } A \cap B}{\# \text{ of elements in } B}$$

ex: A ball is selected from an urn containing 2 black balls, numbered 1 and 2, two white balls, numbered 3 and 4. A is the event that black ball selected and B is the event that even-numbered ball is selected. $P(A|B)$?

$$S = \{ (1,b), (2,b), (3,w), (4,w) \}$$

$$A = \{ (1,b), (2,b) \}$$

$$B = \{ (2,b), (4,w) \}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/2} = \frac{1}{2}$$

$$= \frac{P(\{(2,b)\})}{P(\{(2,b), (4,w)\})} = \frac{1/4}{1/2} = \frac{1}{2}$$

or
equally likely

$$P(A|B) = \frac{\# \text{ of elements in } A \cap B}{\# \text{ of elements in } B} = \frac{1}{2}$$

ex: Radar detection.

- If aircraft is present, radar detects it and generates alarm with prob. 0.99.
- If aircraft not present, radar generates (false) alarm with prob. 0.1.

Aircraft present with prob. 0.05.
what is prob. of no aircraft present, false alarm?
" " of aircraft present, no detection?

Need sequential representation of the experiment by using 2 events.

$$A = \{ \text{aircraft is present} \}$$

$$B = \{ \text{radar generates alarm} \}$$

$$A^c = \{ \text{aircraft not present} \}$$

$$B^c = \{ \text{radar not generate alarm} \}$$

$$P(B|A) = 0.99 \Rightarrow P(B^c|A) = ?$$

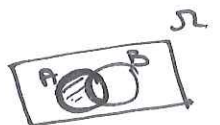
$$= 1 - P(B|A)$$

$$= 1 - 0.99$$

$$= 0.01$$

$$P(B|A^c) = 0.1 \Rightarrow P(B^c|A^c) = 1 - P(B|A^c)$$

$$= 0.9$$



$$P(A) = 0.05 \Rightarrow P(A^c) = 1 - P(A) = 0.95$$

(7)

$$P(A^c \cap B) = ? \quad P(A \cap B^c) = ?$$

Remember

$$P(E_1 | E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} \Rightarrow P(E_1 \cap E_2) = P(E_2) P(E_1 | E_2)$$

$$P(E_2 | E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)} \Rightarrow P(E_1 \cap E_2) = P(E_1) P(E_2 | E_1)$$

$$P(E_1 \cap E_2) = P(E_2) P(E_1 | E_2) = P(E_1) P(E_2 | E_1)$$

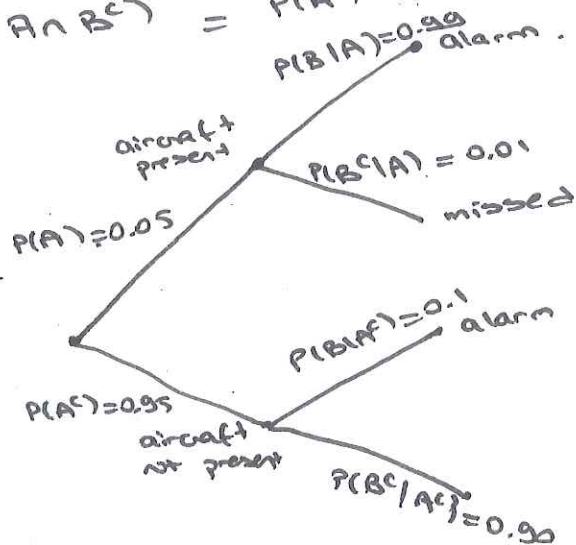
$$\text{OR if } P(E_1 | E_2) = ?$$

$$\text{but know } P(E_2 | E_1) \\ \text{then } P(E_1 | E_2) = \frac{P(E_1) P(E_2 | E_1)}{P(E_2)}$$

Here

$$P(A^c \cap B) = P(A^c) \cdot P(B | A^c) = 0.95 \times 0.1 = 0.095$$

$$P(A \cap B^c) = P(A) \cdot P(B^c | A) = 0.05 \times 0.01 = 0.0005$$

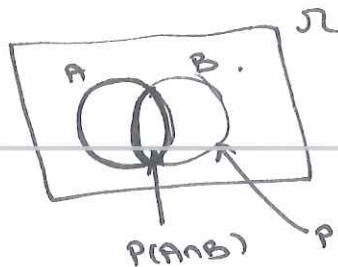


Now, given that radar generates alarm, what is the probability that aircraft is present?

$$P(A | B) = ?$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B | A)}{P(B)} = \frac{0.05 \times 0.99}{P(B)}$$

?



$$B = (A \cap B) \cup (A^c \cap B)$$

↑ ↑
no intersection no intersection

$$P(B) = P(A \cap B) + P(A^c \cap B)$$

$$= P(A) \cdot P(B|A) + P(A^c) \cdot P(B|A^c)$$

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(A) \cdot P(B|A) + P(A^c) \cdot P(B|A^c)}$$

$$= \frac{0.05 \cdot 0.99}{0.05 \cdot 0.99 + 0.95 \cdot 0.1}$$

ex: in a chess tournament, half of the players are type 1, quarter of players are type 2, remaining quarter of players are type 3. Your probability of winning is 0.3 against type 1 players, 0.4 against type 2 players, 0.5 against type 3 players. Against a randomly chosen opponent, what is the probability of winning?

A_i : event of playing against opponent i .

$$P(A_1) = \frac{1}{2}$$

$$P(A_2) = \frac{1}{4}$$

$$P(A_3) = \frac{1}{4}$$

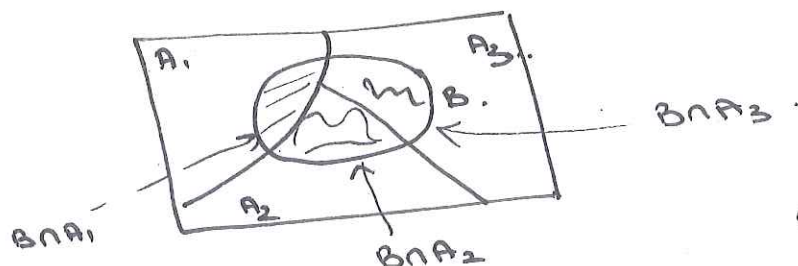
$$A_1 \cap A_2 = \emptyset, A_1 \cap A_3 = \emptyset, A_2 \cap A_3 = \emptyset$$

$$A_1 \cup A_2 \cup A_3 = \Omega$$

B : event of winning

$$P(B|A_1) = 0.3, P(B|A_2) = 0.4, P(B|A_3) = 0.5$$

$$P(B) = ?$$



$$B = (B \cap A_1) \cup (B \cap A_2) \cup (B \cap A_3)$$

↑ ↑ ↑
no intersection no intersection no intersection

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3)$$

$$= P(A_1) P(B|A_1) + P(A_2) P(B|A_2) + P(A_3) P(B|A_3)$$

$$P(B) = \frac{1}{2} \cdot 0.3 + \frac{1}{4} \cdot 0.4 + \frac{1}{4} \cdot 0.5$$

Total Probability Theorem:

Let $A_1, \dots, A_n$ disjoint events that form a partition of sample space ($A_1 \cup A_2 \dots \cup A_n = \Omega$).

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B) \\ = P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + \dots + P(A_n) \cdot P(B|A_n)$$

- Suppose that you win. what is the probability that you had an opponent of type 1?

$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{P(B|A_1) P(A_1)}{P(B|A_1) P(A_1) + P(B|A_2) P(A_2) + P(B|A_3) P(A_3)} \\ = \frac{\frac{1}{2} \cdot 0.3}{\frac{1}{2} \cdot 0.3 + \frac{1}{4} \cdot 0.4 + \frac{1}{4} \cdot 0.5}$$

Bayes' rule:

Let $A_1, \dots, A_n$ disjoint events that form a partition of sample space. then

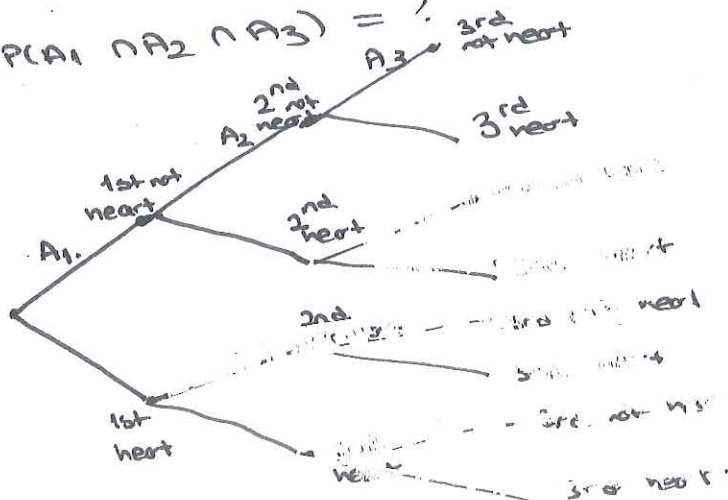
$$P(A_i|B) = \frac{P(A_i) P(B|A_i)}{P(B)} \\ = \frac{P(A_i) P(B|A_i)}{P(A_1) P(B|A_1) + \dots + P(A_n) P(B|A_n)}$$

ex: Three cards drawn from 52-card deck without replacement. what is the probability that one of the cards is a heart?

Define events

$A_i = \{i\text{th card is not a heart}\} \quad i=1,2,3.$

$$P(A_1 \cap A_2 \cap A_3) = ?$$



$$P(A_1) = \frac{39}{52} \quad \leftarrow \begin{array}{l} \text{\# of not} \\ \text{heart} \end{array} \quad \leftarrow \text{total \#}$$

$$P(A_2|A_1) = \frac{38}{51}$$

$$P(A_3|A_1 \cap A_2) = \frac{37}{50}$$

$$\begin{aligned}
 P(\underbrace{A_1 \cap A_2}_{E_1} \cap A_3) &= P(E_1 \cap A_3) = P(E_1) \cdot P(A_3 | E_1) \\
 &= P(A_1 \cap A_2) P(A_3 | A_1 \cap A_2) \\
 &= P(A_1) \cdot P(A_2 | A_1) P(A_3 | A_1 \cap A_2) \\
 &= \frac{39}{52} \cdot \frac{38}{51} \cdot \frac{37}{50}
 \end{aligned}$$

Multiplication rule:

$$P(A_1 \cap A_2 \dots \cap A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_1 \cap A_2) \dots P(A_n | A_1 \cap A_2 \dots A_{n-1})$$

Independence:

$$P(A|B) = P(A)$$

↑
occurrence of B not provide any additional information.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A) \Rightarrow P(A \cap B) = P(A) \cdot P(B)$$

ex: Consider rolling a single die.

$$A = \{1, 3, 5\}$$

$$B = \{2, 4, 6\}$$

Are A and B independent?

$$P(A \cap B) = 0 \Rightarrow A \text{ and } B \text{ are disjoint events.}$$

actually $B = A^c$ given that $S = \{1, 2, 3, 4, 5, 6\}$.

$$P(B|A) = P(A^c | A) = 0.$$

occurrence of A gives complete info about B.

ex: Consider rolling a single die.

$$A = \{1, 3, 5\}$$

$$E = \{1, 2\}$$

Are A and E indep?

$$P(A \cap E) = P(\{1\}) = \frac{1}{6} \stackrel{?}{=} P(A) \cdot P(E) = \frac{2}{6} \cdot \frac{3}{6} \quad \checkmark$$

yes. indep.

ex: Consider rolling a die twice.

$A = \{1^{\text{st}} \text{ roll is a } 1\}$

$B = \{ \text{sum of two rolls is a } 7 \}$

Are A and B independent?

$$P(A \cap B) = P(\{1-6\}) = \frac{1}{36}$$

$$P(A) = P(\{1-1, 1-2, 1-3, 1-4, 1-5, 1-6\}) = \frac{1}{6}$$

$$P(B) = P(\{1-6, 2-5, 3-4, 4-3, 5-2, 6-1\}) = \frac{1}{6}$$

$$P(A \cap B) = P(A) \cdot P(B) \quad \text{Yes.}$$

Conditional Independence:

Two events do not say anything about each other given the third.

$$\begin{aligned} P(A \cap B | C) &= P(A | C) \cdot P(B | C) \\ P(A | B) &= P(A) \\ P(B | A) &= P(B) \end{aligned} \Rightarrow \begin{aligned} P(A | B \cap C) &= P(A | C) \\ P(B | A \cap C) &= P(B | C) \end{aligned}$$

A and B independent $\nRightarrow$ A and B indep. given C.

ex: Consider 2 coin tosses.

$A = \{1^{\text{st}} \text{ toss is head}\}$

$B = \{2^{\text{nd}} \text{ toss is head}\}$

$D = \{ \text{the tosses have different results} \}$

A and B indep?

$$P(A \cap B) = P(\{H-H\}) = \frac{1}{4} \stackrel{?}{=} P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{2} \quad \checkmark$$

A and B indep. given D?

$$P(A \cap B | D) = 0 \stackrel{?}{=} P(A | D) \cdot P(B | D) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \quad \times$$

ex: Two coins: a blue and a red one.

Blue coin: $P(H) = 0.99$

Red coin: $P(H) = 0.01$

Pick a random coin and toss it twice.

$H_1 = \{1^{\text{st}} \text{ toss is head}\}$

$H_2 = \{2^{\text{nd}} \text{ toss is head}\}$

$B = \{ \text{blue coin selected} \}$

Are H_1 and H_2 indep. given B ?

$$P(H_1 \cap H_2 | B) = P(H_1 | B) P(H_2 | B) = 0.99 \times 0.99.$$

Are H_1 and H_2 indep?

$$P(H_1 \cap H_2) = P(B) \cdot P(H_1 \cap H_2 | B) + P(B^c) \cdot P(H_1 \cap H_2 | B^c) \\ = \frac{1}{2} \cdot 0.99 \cdot 0.99 + \frac{1}{2} \cdot 0.01 \cdot 0.01 \approx \frac{1}{2}$$

$$P(H_1) = P(B) P(H_1 | B) + P(B^c) P(H_1 | B^c) \\ = \frac{1}{2} \cdot 0.99 + \frac{1}{2} \cdot 0.01 = \frac{1}{2}.$$

$$P(H_2) = P(B) P(H_2 | B) + P(B^c) P(H_2 | B^c) \\ = \frac{1}{2} \cdot 0.01 + \frac{1}{2} \cdot 0.99 = \frac{1}{2}.$$

$$P(H_1 \cap H_2) \approx \frac{1}{2} \neq P(H_1) \cdot P(H_2) \quad \text{not indep.}$$

Independence of Multiple Events:

$A_1, A_2, \dots, A_n$ are indep. if

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i) \quad \text{for every subset } S \text{ of } \{1, 2, \dots, n\}$$

ex: Three events: A_1, A_2, A_3 .

$$P(A_1 \cap A_2) = P(A_1) P(A_2)$$

$$P(A_1 \cap A_3) = P(A_1) P(A_3)$$

$$P(A_2 \cap A_3) = P(A_2) P(A_3)$$

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) P(A_2) P(A_3).$$

} pairwise indep.

Pairwise indep. DOES NOT IMPLY independence

ex: Consider two fair coin tosses

$H_1 = \{1st \text{ toss is a head}\}$

$H_2 = \{2nd \text{ toss is a head}\}$

$D = \{\text{the two tosses have diff. results}\}$

H_1, H_2 indep.

$$P(D | H_1) = \frac{P(D \cap H_1)}{P(H_1)} = \frac{1/4}{1/2} = \frac{1}{2} = P(D)$$

D, H_1 indep; D, H_2 indep. $\Rightarrow$ pairwise indep.

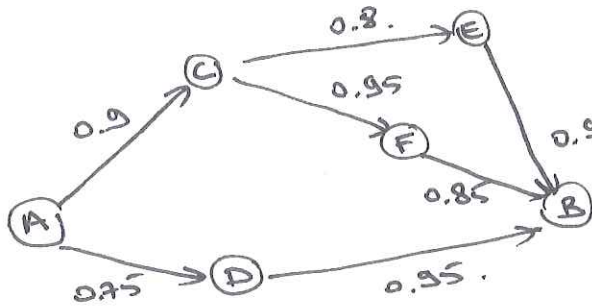
$$P(H_1 \cap H_2 \cap D) = 0 \neq \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = P(H_1) P(H_2) P(D)$$

not indep.

Reliability:

complex systems involving several components,
assume behavior of all components are independent.
→ simplify calculation.

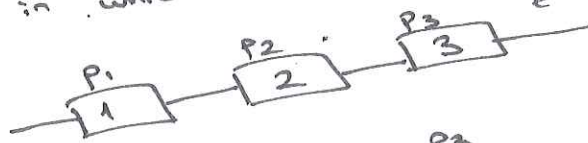
ex: Network connectivity.



probability that comm. (A, B) successful.
link failures indep.

What is the probability that there is a path
connecting A and B in which all links are successful?
← success prob.

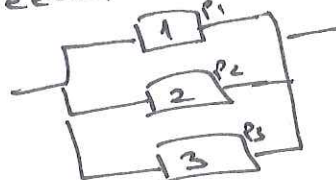
series connection:



$P(\text{series subsystem})$

$$\text{succeeds} = P_1 P_2 P_3$$

parallel connection:



$$P(\text{parallel subsystem succeeds}) = 1 - P(\text{parallel subsystem fails}) \\ = 1 - (1 - P_1)(1 - P_2)(1 - P_3)$$

X → Y: event of connection from x to y.

$$P(C \rightarrow B) = 1 - (1 - P(C \rightarrow E \rightarrow B)) (1 - P(C \rightarrow F \rightarrow B)) \\ = 1 - (1 - P_{CE} \cdot P_{EB}) (1 - P_{CF} \cdot P_{FB}) \\ = 1 - (1 - 0.8 \cdot 0.9) (1 - 0.95 \cdot 0.85) \\ = 0.946$$

$$P(A \rightarrow C \text{ and } C \rightarrow B) = P(A \rightarrow C) P(C \rightarrow B) = 0.9 \cdot 0.946 = 0.851$$

$$P(A \rightarrow D \text{ and } D \rightarrow B) = P(A \rightarrow D) P(D \rightarrow B) = 0.75 \cdot 0.95 = 0.712$$

$$P(A \rightarrow B) = 1 - (1 - P(A \rightarrow C \text{ and } C \rightarrow B)) (1 - P(A \rightarrow D \text{ and } D \rightarrow B)) \\ = 1 - (1 - 0.851) (1 - 0.712) \\ = 0.957$$

Independent trials and Binomial probabilities;

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A seqn. of independent but identical stages
→ independent trials.

Each trial has two outcomes: failure or success
→ Bernoulli trials.

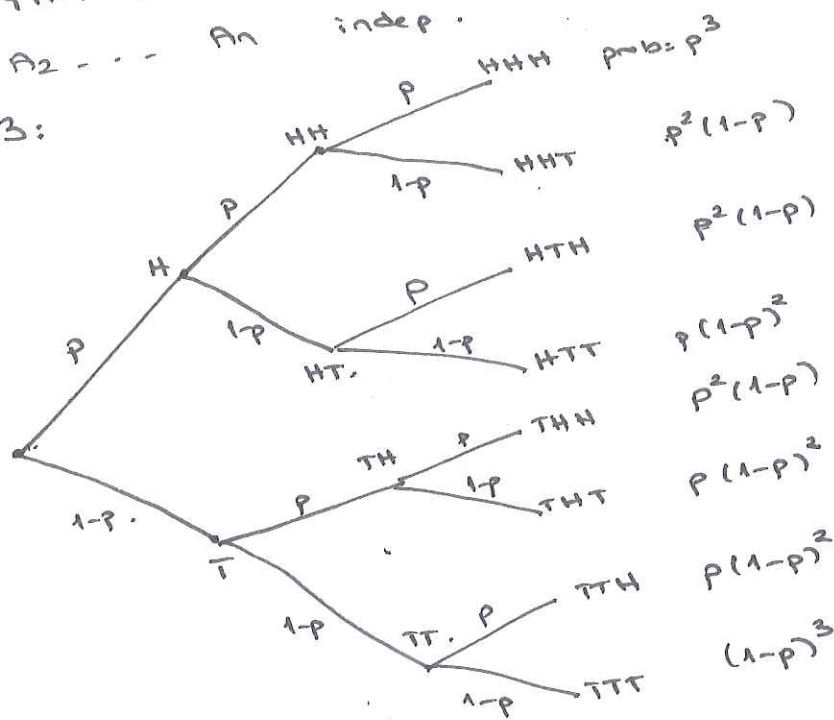
n independent coin tosses.

$A_i = \{ \text{ith toss is a head} \}$.

$$P(A_i) = p.$$

$A_1, A_2, \dots, A_n$ indep.

$n=3$:



$$P(\{HHH\}) = p^3$$

$$P(\{HTH\}) = p^2(1-p).$$

$$P(2 \text{ heads one up in 3-tosses}) = p^2(1-p) + p^2(1-p) + p^2(1-p) = 3 p^2(1-p)$$

↑
of outcomes with 2 heads

$$P(k \text{ heads in } n\text{-tosses}) = \binom{n}{k} p^k (1-p)^{n-k}$$

where $i! = 1 \cdot 2 \cdot \dots \cdot (i-1)!$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Counting:

(15)

1) If sample space Ω has a finite # of equally likely outcomes,

$$P(A) = \frac{\# \text{ of elements in } A}{\# \text{ of elements in } \Omega}.$$

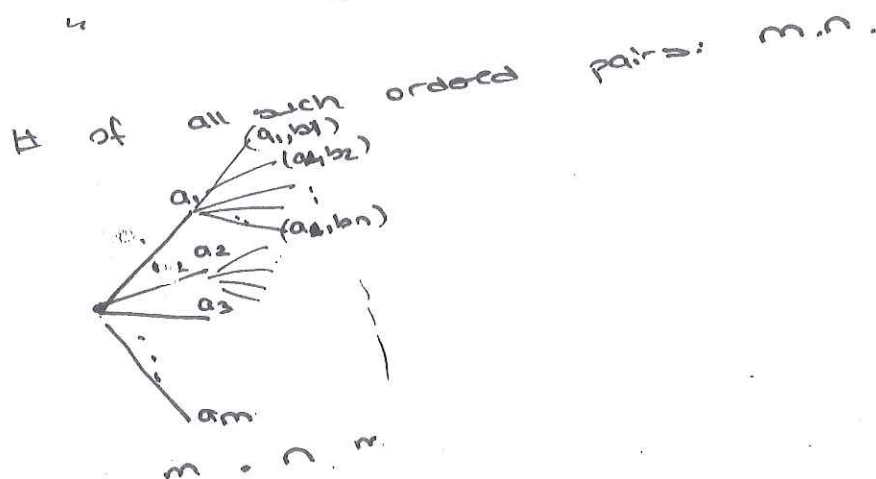
2) If A contains a finite # of equally likely outcomes, each with prob. p .

$$P(A) = p \cdot (\# \text{ of elements in } A)$$

ex: k -head in n -^{coin} tosses.

Counting principle:

An experiment with two stages.
 possible outcomes for first stage: $a_1, a_2, \dots, a_m$
 " " " " second " : $b_1, b_2, \dots, b_n$
 " " " " for experiment: (a_i, b_j) , $i=1, 2, \dots, m$
 $j=1, 2, \dots, n$.



A process with r stages:

n_i possible result at i -th stage.

Total # of possible results: $n_1 \cdot n_2 \cdot \dots \cdot n_r$.

ex: # of telephone numbers

Phone number: 7-digit seqn.

First digit cannot be a 1 or 0

How many distinct phone numbers?

7-stages.

First stage: 2, 3, ..., 9 : 8 options

2nd, 3rd, ..., 7th stage: 0, 1, ..., 9: 10 options

$$8 \times \underbrace{10 \times 10 \times \dots \times 10}_{6 \text{ times}} = 8 \cdot 10^6$$

Selection of k objects out of n objects:

If order matters $\Rightarrow$ permutation

Otherwise $\Rightarrow$ combination

Permutation:

n distinct objects
pick k out of n objects and arrange them in sequence.
how many different seqs. of k objects?

of options for first: n .
" " " for 2nd: $n-1$.

" " " k th: $n-(k-1)$

of different seqs. of k objects: $n(n-1) \dots n-(k-1)$
 $= \frac{n(n-1)(n-2) \dots (n-k+1)(n-k) \dots 1}{(n-k) \dots 1}$

$$= \frac{n!}{(n-k)!}$$

ex: # of words that consists of four distinct letters.
26 letters in alphabet.

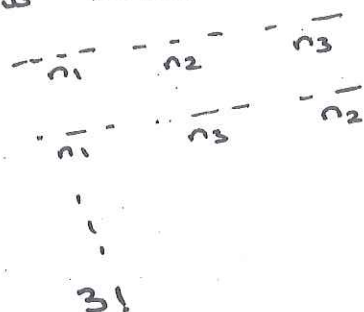
$$26 \cdot 25 \cdot 24 \cdot 23$$

$$\text{H-permutations of 26 letters: } \frac{26!}{22!} = 26 \cdot 25 \cdot 24 \cdot 23$$

ex: n_1 : # of classical CDs
 n_2 : # of rock CDs

n_3 : # of crate CD's.

keep all CDs of one kind together.
How many ways to arrange them?



within each: $n_1! \cdot n_2! \cdot n_3!$

$$3! \cdot n_1! \cdot n_2! \cdot n_3!$$

Combination:

n people, select a committee of k .

How many different committees?

In a combination, no ordering.

2 permutations of letters A, B, C, D.

$\{AB, BA, AC, CA, AD, DA, BC, CB, BD, DB, CD, DC\}$.

2 combinations of letters A, B, C, D:

$\{AB, AC, AD, BC, BD, CD\}$

each combination is associated with $k!$ duplicate k -permutations.

of possible combinations: $\frac{n!}{k!(n-k)!} = \binom{n}{k}$

ex: A production lot of size 100 known to have 5 defective items.
A random sample of 10 selected without replacement

a) what is the probability that sample has no defectives?

$$\frac{\binom{95}{10}}{\binom{100}{10}}$$

b) what is the prob. sample has all defectives?

$$\frac{\binom{5}{5} \binom{95}{5}}{\binom{100}{10}}$$

Partitioning:

combination: choice of k elements out of n -element set
 $\equiv$ partition of a set into two:

one set: k elements

other set: $(n-k)$ elements

partition of a set into r subsets: $n_1, n_2, \dots, n_r$
st. $n_1 + n_2 + \dots + n_r = n$.

How many ways this can be done?

$$\begin{aligned} & \binom{n}{n_1} \binom{n-n_1}{n_2} \binom{n-n_1-n_2}{n_3} \dots \binom{n-n_1-\dots-n_{r-1}}{n_r} \\ &= \frac{n!}{n_1!(n-n_1)!} \cdot \frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \dots \frac{(n-n_1-\dots-n_{r-1})!}{n_r!} \\ &= \frac{n!}{n_1! n_2! \dots n_r!} = \binom{n}{n_1, n_2, \dots, n_r} \end{aligned}$$

ex: How many diff. letter seqs. can be obtained by rearranging 2 letters in the word TATTOO?

6 positions to be filled.

group of size 3:

1st group positions for letter T
2nd " " " " A
3rd " " " " O.

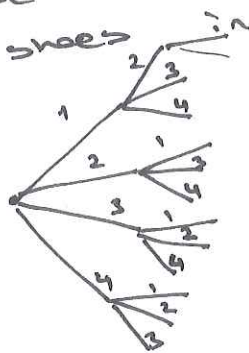
$$\frac{6!}{1! 2! 3!}$$

Examples:

17. A student has 4 different pairs of shoes. In how many ways can he wear shoes in 5 days?

$$4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 = 4^5$$

If he never wears the same pair on two consecutive days, in how many ways can he wear shoes in 5 days?



$$4 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 4 \cdot 3^4$$

27 In how many ways can 10 students occupy 10 desks?

$$10!$$

In how many ways can 10 students occupy 12 desks?

$$\binom{12}{10} 10!$$

37 You win a lottery if you correctly predict numbers of six balls drawn from an urn containing balls numbered 1, 2, ..., 49, without replacement and without regard to ordering. What is the probability of winning if you buy one ticket?

$$\frac{1}{\binom{49}{6}}$$

4) Problem 16:

We are given 3 coins: one has heads in both faces, second has tails in both faces, third has head in one face and tail in the other. Choose a coin at random, toss it and result is head. What is the prob. opposite face is tail?

$$P(\text{third coin} | \text{head come up})$$

$$= \frac{P(\text{third coin and head come up})}{P(\text{head come up})}$$

$$= \frac{P(\text{head come up} | \text{third coin}) P(\text{third coin})}{P(\text{head come up})}$$

$$P(\text{head come up} | \text{third coin}) = \frac{1}{2}.$$

$$P(\text{third coin}) = \frac{1}{3}.$$

$$P(\text{head come up}) = P(\text{head come up} | \text{first coin}) P(\text{first coin}) + P(\text{head} | \text{second coin}) P(\text{second coin}) + P(\text{head} | \text{third coin}) P(\text{third coin})$$

$$= 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{3}$$

$$= \frac{1}{2}$$

$$= \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}.$$

5) problem 46:

Laplace's rule of succession

Consider $m+1$ boxes with k -th box containing k red balls and $(m-k)$ white balls, where k ranges from 0 to m . Choose a box at random and then choose a ball at random from the same box n successive times (ball is replaced each time). Suppose a red ball drawn each of the n times. What is the probability that if we draw a ball one more time, it will be red?

E : event of a red ball drawn at time $n+1$
 R_n : event of a red ball drawn each of the n preceding times.

(20)

$$P(E|R_n) = \frac{P(E \cap R_n)}{P(R_n)}$$

$$P(R_n) = P(\text{1st box chosen}) \cdot \left(\frac{0}{3}\right)^n + P(\text{2nd box chosen}) \left(\frac{1}{3}\right)^n + \dots + P(\text{(n+1)th box}) \left(\frac{n}{n}\right)^n$$

$$= \frac{1}{n+1} \sum_{k=0}^n \left(\frac{k}{n}\right)^n$$

$$P(E \cap R_n) = P(R_{n+1}) = \frac{1}{n+1} \sum_{k=0}^n \left(\frac{k}{n}\right)^{n+1}$$

$$P(E|R_n) = \frac{\sum_{k=0}^n \left(\frac{k}{n}\right)^{n+1}}{\sum_{k=0}^n \left(\frac{k}{n}\right)^n}$$

large n , $P(E|R_n) \approx 1$